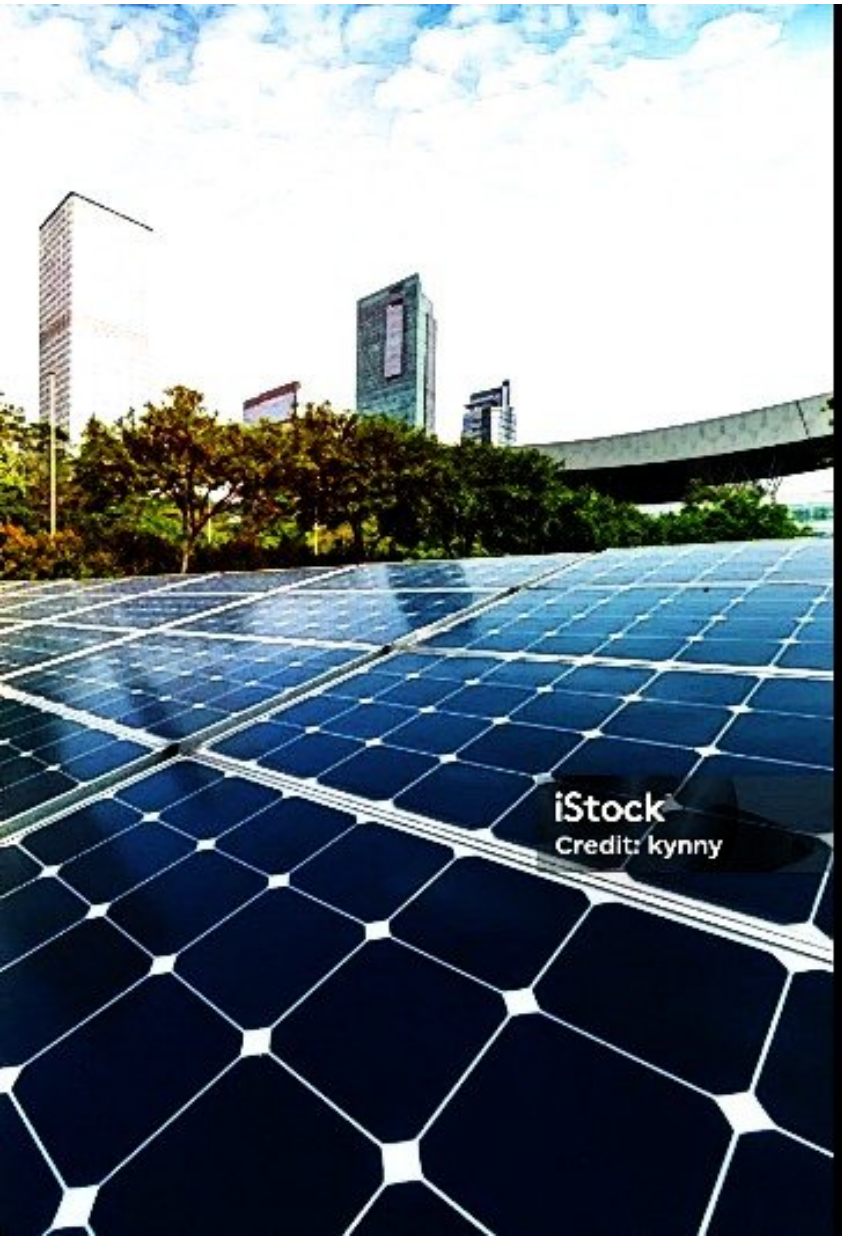

Energy Consumption Analysis: Optimising for a Sustainable Future

Agenda: Our Journey Through Energy Insights

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Why It Matters

Impact on cost, environment, and operational efficiency

02

Current Footprint

Data collection and baseline establishment

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What to measure and why it's important

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From smart meters to artificial intelligence

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Identify Waste

Case studies and common pitfalls

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Reduction Strategies

Practical steps for immediate impact

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Renewables

Future of your energy mix

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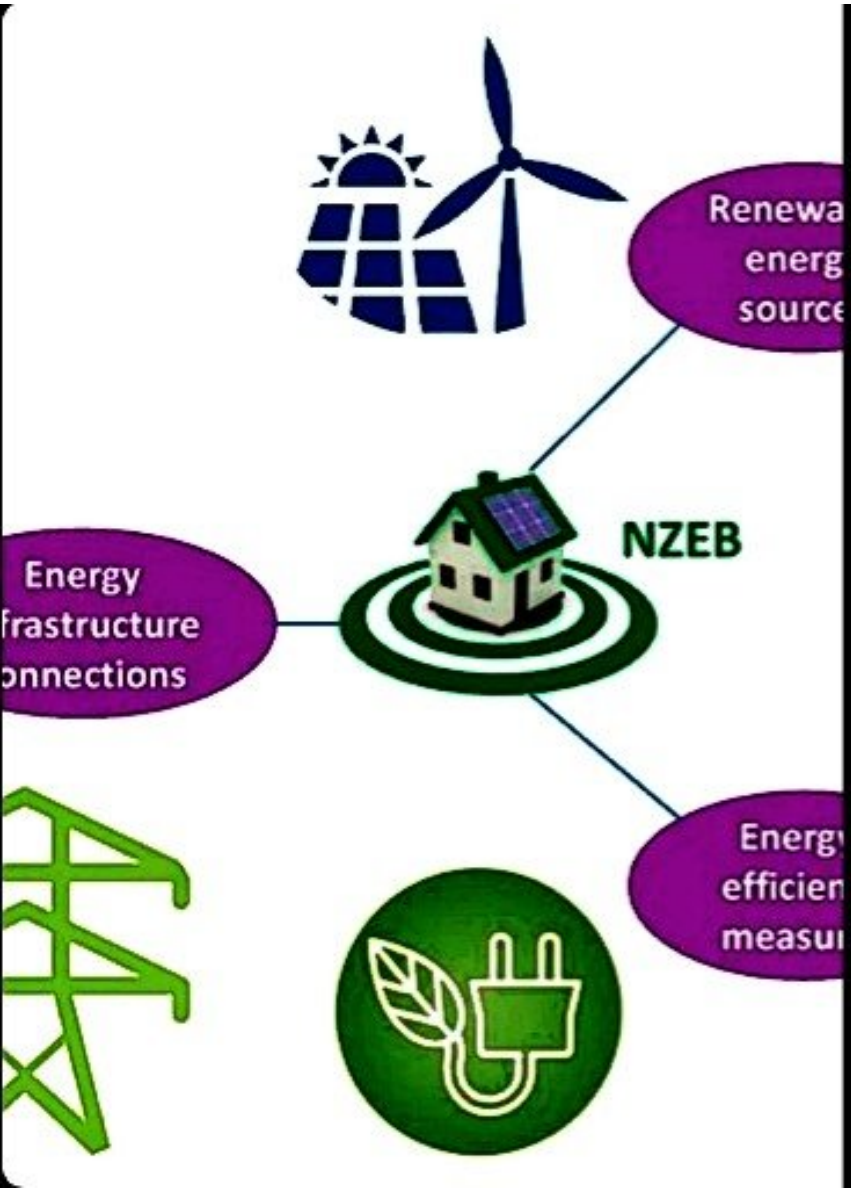
Next Steps

Your path to optimisation



Why Energy Analysis Matters

Cost Reduction

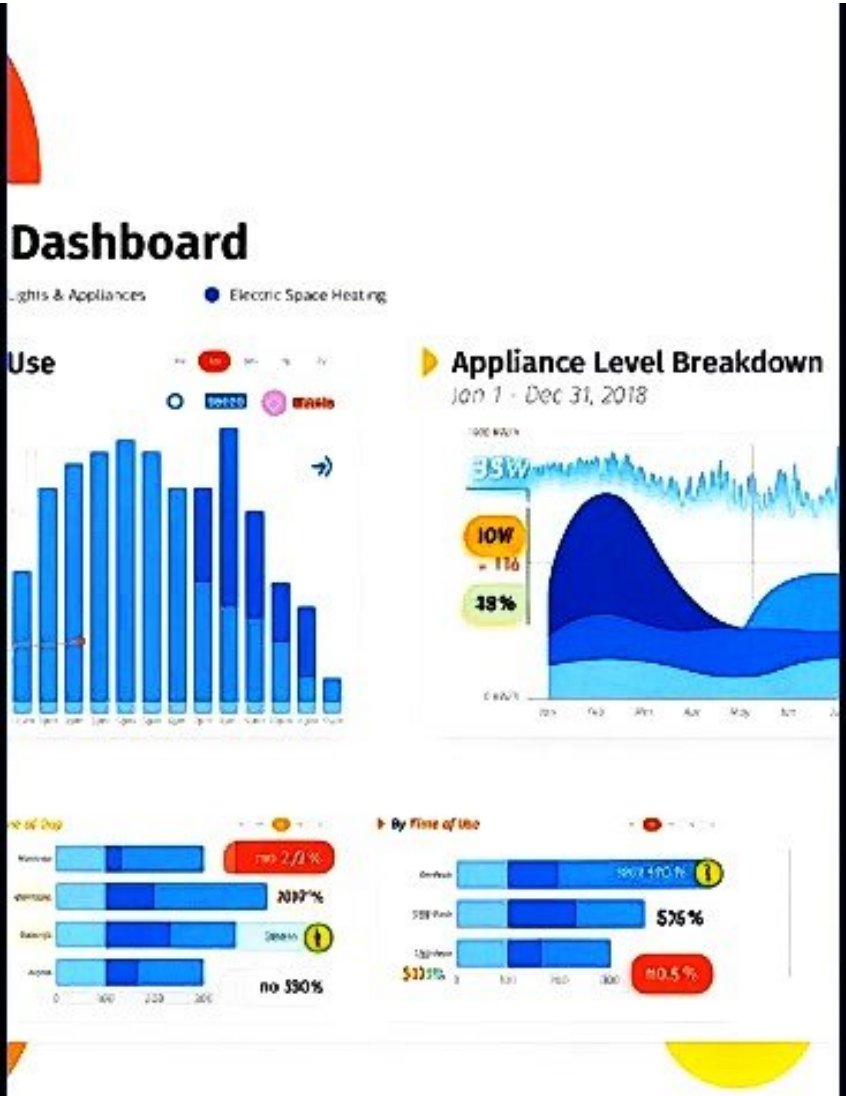
Energy typically represents 15-30% of operational expenses. Analysis identifies waste, leading to 20-40% savings potential.

Environmental Impact

Buildings account for 40% of global carbon emissions. Reducing consumption directly lowers your carbon footprint.

Operational Efficiency

Optimised systems perform better, reducing downtime and extending equipment lifespan through proper maintenance.



Understanding Your Current Energy Footprint

Data Collection

Begin with utility bills covering 12-24 months to establish consumption patterns. Install sub-meters for granular tracking across departments or equipment.

- Electricity, gas, water usage
- Time-of-use patterns
- Seasonal variations
- Peak demand periods

Baseline Establishment

Calculate baseline consumption per unit area (kWh/m²) or per unit of production. This benchmark enables progress tracking and goal setting.

- Normalise for weather conditions
- Account for occupancy changes
- Set realistic reduction targets

Key Metrics and Indicators

Measure what matters to drive improvements and track progress.

Energy Intensity

Energy used per square metre or unit produced. It helps benchmark how efficiently a site operates.

Peak Demand

The highest power draw at any time. Lowering peaks can reduce demand charges and ease grid stress.

Load Factor

The ratio of average to peak demand. Higher values mean steadier, more efficient energy use.

Carbon Footprint

Emissions linked to energy use. Tracking it supports progress toward lower-carbon and net-zero goals.

Tools and Technologies for Energy Analysis

Smart Meters

Track energy use in real time with granular interval data to reveal waste and anomalies.

IoT Sensors

Monitor equipment conditions to detect inefficiencies early and prevent failures.

Energy Management Systems

Centralise data from multiple sources and automate reporting, alerts, and dashboards.

AI Analytics

Use machine learning to predict consumption, optimise schedules, and recommend actions.

Identifying Waste & Inefficiencies

Common Pitfalls

- **Vampire loads:** Standby power accounts for 5-10% of residential consumption, higher in commercial settings
- **Over-ventilation:** Excessive fresh air brings heating/cooling loads, wasting 15-20% of HVAC energy
- **Temperature setbacks:** Inconsistent settings force systems to work harder, increasing consumption 2-5% per degree
- **Equipment ageing:** Motors and compressors lose 1-2% efficiency annually without maintenance

Case Study: Manufacturing Plant

Analysis revealed compressed air leaks costing ₹18 lakh annually. Repair investment: ₹4.5 lakh. Payback period: 3 months. Annual savings: ₹18 lakh.

Strategies for Reduction & Optimisation

Practical actions to cut energy use, lower costs, and improve efficiency.

1

Lighting Upgrades

Switch to LEDs for 50–70% savings. Add sensors for fast ROI in 1–3 years.

2

HVAC Optimisation

Use maintenance, insulation, and smart controls to cut heating/cooling by 20–40%.

3

Equipment Scheduling

Move non-essential loads off-peak and automate shutoffs during idle periods.

4

Employee Engagement

Training and awareness can deliver 5–15% savings through simple daily actions.

Thank You...